MIL-S-55648B(EL)
5 September 1972
Superseding
MIL-S-55648A(EL)
9 September 1970

MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT S-393( )/TRC-138

1. SCOPE

1.1 Scope.- This specification covers the air or vehicular transportable shelter facility part of the Repeater Set, Radio AN/TRC-138( ) which is capable of providing multiplex repeater facilities for 48 or 96 channels of voice frequency communications for radio or cable transmission. (See 6.1 and 6.5)

1.2 Description.- Shelter, Electrical Equipment S-393( )/TRC-138 hereinafter called the shelter facility, includes a modified S-280( )/G Shelter and all racks, brackets, shelves, tie-downs, electrical components and wiring to enable installation and operation of the assemblage components of the AN/TRC-138( ). (See 6.1 and 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)
MIL-W-6858	Welding, Aluminum, Magnesium, Non-Hardening Steels or Alloys and Titanium, Spot, Seam and Stitch
MIL-T-7928	Terminals, Lug and Splices, Crimp Style, Copper

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MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-W-45205	Welding, Gas, Metal-Arc and Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable, for Structures, Excluding Armor
MIL-P-55505	Shelter, Electrical Equipment (Containing Equipment) Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-1235	Single, Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ELECTRONICS COMMAND

SC-A-46691	RF Cable Assembly Test
DL-SC-B-632352	Shelter, Electrical Equipment S-393( )/TRC-138
SC-D-632457	Power Schematic Diagram S-393( )/TRC-138
SC-F-632468	Power Wiring Diagram S-393( )/ TRC-138
SC-D-632533	Sighting Cable
SC-D-632535	Equipment Interconnect Harness
SC-D-632538	Signal Schematic
SC-D-632645	Indicator Box Schematic Diagram

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(Copies of specifications standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 First article.- When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.- The shelter facility shall be constructed in accordance with drawings as listed on DL-SC-B-632352. (See 4.6 and 3.22)

3.3 Electrical installation.- (See 3.22)

3.3.1 Power systems and components installation.- (See 4.7.1) Installation of power wires and components (including cable assemblies, receptacles, switches, meters, circuit breakers, neon lamps, fluorescent lamps, incandescent lamps, lighting fixtures, distribution assembly system, and wiring components) constituting a three wire electrical power distribution system, shall be installed and wired in accordance with the drawings. The complete in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of conductors.- The point-to-point resistance of all power-line conductors shall be less than 3 ohms when tested in accordance with 4.7.1.1.

3.3.1.2 Dielectric breakdown voltage.- The dielectric strength of insulation of all power-line wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 30 seconds when tested in accordance with 4.7.1.2.

3.3.1.3 Insulation resistance.- The resistance of insulation between any two of the three power-line conductors and also between each of the two ungrounded power-line conductors and shelter facility ground terminal shall not be less than 100 megohms when tested in accordance with 4.7.1.3.

3.3.2. Power distributional and operational functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as shown by Power Schematic Drawing SC-D-632457, when tested in accordance with 4.7.2.

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3.3.3 Signal (audio) system wiring and components installation.- (See 4.7.3) Installation of signal (audio wires, cables and wiring components including binding posts, connectors, plugs, jacks, receptacles, and wiring devices) shall be installed and wired in accordance with the drawings. The completed in-place signal system installation shall meet the following requirements.

3.3.3.1 Continuity of conductors.- The resistance between both end terminations of all signal-line conductors shall not exceed 5 ohms when tested in accordance with 4.7.3.1.

3.3.3.2 Dielectric and insulation resistance tests.- The insulation of all signal wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 5 seconds and the insulation resistance shall not be less than 100 megohms when tested in accordance with 4.7.3.2.

3.3.4 Video and radio frequency (RF), system wiring and components installation.- (See 4.7.4) Installation of video and RF cable assemblies, waveguides, and connecting components (including plugs, connectors, adapters, fittings, receptacles, and lightning protectors) shall be in accordance with the drawings. The completed in-place video and RF system installation shall meet the following requirements:

3.3.4.1 Continuity of conductors.- The point-to-point resistance between the following video and RF cable Terminations shall not exceed 3 ohms when tested in accordance with 4.7.4.1:

a. Both end terminations of the inner conductors.

b. Both end terminations of the outer conductor (shield).

3.3.4.2 Dielectric and insulation resistance.- The insulation of all video and RF cables, components and lighting protectors shall withstand the following potentials with the insulation resistance not less than 100 megohms when tested in accordance with 4.7.4.2:

a. 1500 volts DC applied continuously for not less than 30 seconds to all video cable assemblies terminated at the video entrance box with lightning protector elements removed.

b. 1100 volts DC applied continuously for not less than 30 seconds to all video cable assemblies terminated at the video entrance box with lightning protector elements in place.

c. 1500 volts DC applied continuously for not less than 5 seconds to all RF cable assemblies inside shelter facility.

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3.3.4.3 Insertion Loss.- The insertion loss (forward power transmission loss) between any two adjacent ports, with the third port terminated, shall not exceed 1.25 db anywhere in the frequency range 4.4 to 5.0 gigahertz when tested in accordance with 4.7.4.3.

3.3.4.4 Voltage Standing Wave Radio (VSWR).- The VSWR of all waveguide assemblies in place shall not exceed 1.20 anywhere in the frequency range 4.4 to 5.0 gigahertz when tested in accordance with 4.7.4.4.

3.3.4.5 Isolation.- The waveguide system in place shall provide greater than 20 db isolation over the frequency range 4.4 to 5.0 gigahertz when tested in accordance with 4.7.4.5.

3.3.5 Cable assemblies.- (See 4.4)

3.3.5.1 Cable assemblies, power and audio frequency.- All loose power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings and meet the continuity, dielectric, insulation resistance, and pull tests and requirements of MIL-C-3885.

3.3.5.2 Cable assemblies, radio frequency.- Radio frequency cable assemblies required for operation of major items of equipment in 6.1 shall be fabricated in accordance with the drawings and meet the requirements and tests as indicated in Drawing SC-A-46691.

3.3.6 Suppression of electromagnetic interference.- The motors in the ventilating equipment used in the facility shall meet the emanation limits of MIL-STD-461 for tactical equipment. (See 4.4)

3.4 System operation.- With all equipment in the AN/TRC-138( ) assemblage installed and with the external multiplexers and combiners positioned so as to facilitate testing, the operational requirements of all audio and RF cable assemblies, waveguides, patch panels, and indicator boxes, shall be met when tested in accordance with 4.8.

3.5. Vehicular transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicles without sustaining any permanent damage such as buckling, crackings, delamination, or other permanent deformation. (See 4.9)

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3.6 Rail transportaion.- The shelter facility, without any extra tieing, taping, etc., shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail and shall be subjected to and withstand the test of paragraph 4.10 without sustaining any buckling, cracking, delamination, or other permanent deformation.

3.7 Drops, flat and rotational.- The shelter facility shall be capable of withstanding drops of 18 inches onto concrete. After all drop tests of 4.11.1 and 4.11.2 there shall be no buckling, cracking, delamination or other permanent deformation.

3.8 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.12. During and after the test there shall be no leakage into the shelter facility or its walls, floor, roof, or door.

3.9 Fording.- The shelter facility shall be capable of withstanding the tests of 4.13. During and after the test there shall be no leakage into the shelter facility or its walls, floor, or door.

3.10 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive, and when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.14. There shall be no damage or permanent distortion to the shelter facility mounting surface or insert head or threads and the inserts shall not have rotated from its original position. Insert threads shall be free of paint.

3.11 Cleaning.- (See 3.22)

3.11.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.11.2 Units.- After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware, and other foreign material.

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3.12 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. - (See 4.4)

3.13 Marking.- (See 3.22 and 4.4)

3.13.1 General.- Marking shall conform to Specification MIL-M-13231 and the drawings.

3.13.2 Nameplates.- Each shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.13.3 Aircraft loading data plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the centers-of-gravity of the shelter facility and the assemblage in the three reference planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The locations of the centers-of-gravity shall be determined by the contractor using the first article.

3.13.4 Interior and exterior marking.- All interior and exterior making of the shelter facility shall be as depicted on the drawings.

3.14 Riveting.- (See 3.22) Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.15 Seal, vapor- (See 3.22) All joints and raw edges of the framing for the power, signal, video, and antenna entrance boxes, air-conditioner, and exhaust blowers shall be sealed to prevent entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be one part Silicone Construction Sealant SE-1204 as made by General Electric Company or Silastic 732 as made by Dow Corning, or equal, using the manufacturer recommended primer.

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3.16 Securing of parts.- (See 3.22) Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests.

3.16.1 Screws, threaded devices, and related parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.

3.17 Soldering.- (See 3.22)

3.17.1 Solder and flux.- Solder and flux shall be as defined on the drawings.

3.17.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.17.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, over-heated or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.18 Tropicalization of material.- (See 3.22) Canvas, webbing, wood, etc., shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows:

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3.18.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.18.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.18.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.18.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (see 6.2e)

3.19 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finish weld will be unnecessary except where grinding is specified on the drawings. Inert-gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 Class B and MIL-W-45205 Class B as applicable; surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 3.22 and 4.4)

3.20 Wiring and cabling.- (See 3.22) Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

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3.20.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

b. To facilitate field repair of broken or cut wires.

c. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.20.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.20.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.20.4 Solderless terminal lug connections.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928. (See 4.15)

3.20.5 Identification of wiring.- Wiring shall be identified by color-coding and wire pair marking, as indicated on the drawings

3.21 Interchangeability.- Like units, assemblies, sub-assemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.16) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.22 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs: (See 4.17)

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3.2 Construction
3.3 Electrical installation
3.10 Mounting inserts
3.11 Cleaning
3.12 Finish
3.13 Marking
3.14 Riveting
3.15 Seal, vapor
3.16 Securing of parts
3.17 Soldering
3.18 Tropicalization
3.19 Welding
3.20 Wiring and cabling

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.18)

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4.3 First article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.1)

4.3.1 First article units.- The contractor shall furnish one (1) first article unit of the complete S-393( )/TRC-138.

4.3.2 First article inspection.- The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First article data.- The first article test plan and test report(s) shall be as required in the contract or purchase order.

TABLE I First Article Inspection

Inspections	Rqr Para	Test Para
1. Inspection covered by subsidiary documents	-	4.4
2. Interchangeability	3.21	4.16
3. Mounting Insert	3.10	4.14.1
4. Solderless-terminal-lug connectors	3.20.4	4.15
5. Visual and Mechanical	3.22	4.17
6. Power systems and components installation	3.3.1	4.7.1
7. Power distributional and operational function	3.3.2	4.7.2
8. Signal (audio) system wiring and components installation	3.3.3	4.7.3
9. Video and radio frequency (RF), system wiring and components installation	3.3.4	4.7.4
10. System operation	3.4	4.8
11. Vehicular transportation	3.5	4.9
12. Rail transportation	3.6	4.10
13. Flat drop	3.7	4.11.1
14. Rotational drop	3.7	4.11.2
15. Watertightness	3.8	4.12

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Table I First Article Inspection (continued)

Inspections	Rqr Para	Test Para
16. Fording	3.9	4.13
17. Repeat 6. Power systems and components installation	3.3.1	4.7.1
18. Repeat 7. Power distributional and operational function	3.3.2	4.7.2
19. Repeat 8. Signal (audio) system wiring & components installation	3.3.3	4.7.3
20. Repeat 9. Video and radio frequency (RF) system wiring and components installation	3.3.4	4.7.4
21. Repeat 10. System operation	3.4	4.8

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Cable assemblies	3.3.5
Suppression, RFI	3.3.6
Finish	3.12
Marking	3.13
Welding	3.19

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Lots shall be formed only from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

TABLE II - Group A Inspection

Inspection	Rqr Para	Insp Para	AQL % Def Major	AQL % Def Minor
Visual and Mechanical	3.22	4.17	1.0	4.0
Electrical installation				**
Power systems and component installation				
Continuity of conductors	3.3.1.1	4.7.1.1		
Dielectric breakdown voltage	3.3.1.2	4.7.1.2		
Insulation resistance	3.3.1.3	4.7.1.3		
Power distributional and operational function	3.3.2	4.7.2 thru 4.7.2.3		
Signal (audio) system wiring and components installation				
Continuity of conductors	3.3.3.1	4.7.3.1		
Dielectric and insulation resistance	3.3.3.2	4.7.3.2		
Video and radio frequency (RF), system wiring and components installation				
Continuity of conductors	3.3.4.1	4.7.4.1		
Dielectric & insulation resistance	3.3.4.2	4.7.4.2		
Insertion Loss	3.3.4.3	4.7.4.3		
Voltage Standing Wave Ratio (VSWR), Waveguide transmission lines	3.3.4.4	4.7.4.4		
Isolation	3.3.4.5	4.7.4.5		

1.0 for the entire group of power, signal and video wiring

** All defects are major

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4.5.1.1 Order of inspection within Group A.- Group A inspection shall be performed in the order listed in Table II.

4.5.2 Group B inspection.- Group B inspection shall normally be performed on inspection lots that have passed Group A inspection, except for Interchangeability and Mounting Insert Inspections which are performed prior to Group A. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

TABLE III - Group B Inspection

Inspection	Rqr Para	Insp Para	AQL % Def (See 4.5.2.1)
Interchangeability	3.21	4.16	4.0
Mounting insert	3.10	4.14.2	4.0
Watertightness	3.8	4.12	2.5
Fording	3.9	4.13	2.5

NOTE: The order of inspection of Group B with respect to Groups A and C shall conform to the order specified in 4.3.2. The order of inspection within Group B shall be as in 4.5.2.2.

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in Table II shall be as follows:

AQL	Inspection level
4.0%	S-4
2.5%	S-4

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed on the same equipment in the order in which they appear.

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV. Samples shall be selected in accordance with 4.5.3.1 thru 4.5.3.4.

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TABLE IV - Group C Inspection

Inspection	Rqr Para	Insp Para
Construction verification	3.2	4.6
Solderless-terminal-lug connection	3.20.4	4.15
Mounting inserts	3.10	4.14.1
System operation	3.4	4.8
Vehicular Transportation	3.5	4.9
Rail Transportation	3.6	4.10
Drops, Flat and Rotational	3.7	4.11

NOTE: The order of inspection of Group C with respect to Groups A and B shall conform to the order specified in 4.3 The order of inspection within Group C for all inspections listed shall be as listed in Table IV. Units subjected to Group C Drop Tests shall subsequently be subjected to and meet the watertightness and fording tests of 4.12 and 4.13.

4.5.3.1 Sampling for inspection of construction evaluation.- The Equipment Verification Review, in accordance with paragraph 4.6, shall be performed once on each part and assembly, randomly selected, and on their installation into one of first 10 shelter facilities produced. This inspection shall be performed as the selected unit is being assembled.

4.5.3.2 Sampling for inspection of solderless-terminal-lug-connections.- Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a one foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Sampling for inspection of drop tests.- One sample, representative of the first 10 units or fraction thereof produced, shall be subjected to the drop tests. This sample shall be the same unit previously subjected to the vehicular and rail transportation tests of 4.5.3.4. Thereafter, one sample from each additional 100 units or fraction thereof produced shall be subjected to the drop tests.

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4.5.3.4 Sampling for other inspection.- For inspection tests listed below, one unit shall be selected at random from the first 10 units or fraction thereof which have been produced, unless otherwise specified in the contract. Selection of the unit will be made by the government.

a. Mounting inserts
b. System operation
c. Truck transportability
d. Railroad transportability

4.5.3.5 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the result of investigation of details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of production which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative.

After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed; at the option of the government) and Groups A and B inspection may be reinstituted; final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (See 6.4).

4.5.4 Reinspection of conforming Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage. (See 6.4)

4.5.5 Alternate Group A audit and Group B inspection procedures.- The contractor may elect to use a continuous sampling and inspection plan instead of the lot inspection plans from MIL-STD-105. If this alternate is selected the procedure of implementation must be in accordance with MIL-STD-1235, "Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes". Sampling plans for the AQL's listed in Table II Group A inspection, and Table III Group B inspection of this specification shall be selected from Tables VIII, IX and X of MIL-STD-1235 which are for f=1/2. The number of units in the

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Production Interval shall be the entire production quantity unless otherwise specified in the contract. Before starting production contractor shall notify the contracting officer in writing which method he intends to use. The method may not be changed during production without written approval of the government.

4.6 Construction verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings (see 3.2) cited in the Technical Data Package (as supplemented by all approved engineering changes). The EVR shall consist of the following:

a. An audit to establish that "as-built" shelter facilities, including individual parts, their use in subassemblies, and assemblies (components), and their final installation into the shelter forming the shelter facility, are in accordance with the end product drawings.

b. An audit to establish that "as-built" shelter facilities, including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package for each shelter part, components, and shelter facility. The audit shall record all acceptance test methods used together with resulting verification test data.

c. Preparation of EVR report(s) containing the following information:

(1) Identification of parts, components, shelter facilities and details proving compliance with 4.6a and b above.

(2) Discrepancies noted.

(3) Corrective action taken.

d. Two (2) copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the government technical organization cited in the contract. The results of the EVR shall be subject to verification by authorized government personnel at the time the first unit or production lot is offered for acceptance. Government Verification will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to government personnel during the verification. The contractor shall provide the following:

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(1) Segregation of parts, components of shelter facilities, to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the shelter facility necessary to permit government review of the shelter facility, its components and their installation, and to reassemble parts after inspection.

(2) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

4.7 Electrical installation tests.- The tests of 4.7.1 through 4.7.4.3 shall be performed on each shelter facility. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. Power to be supplied shall be 115 volts AC, 60 cycles, single phase.

4.7.1 Power systems and components installation tests.- (see 3.3.1) Prior to conducting power wiring and components tests of 4.7.1.1, 4.7.1.2, and 4.7.1.3, the following shall be accomplished. (Refer to Drawings SC-D-632457 and SC-F-632468).

a. Remove or disconnect the following electrical loads:

(1) All fluorescent, incandescent, and neon (power indicator and power distribution box) lamps.

(2) All fluorescent lamp starters.

(3) Blowers and heaters.

(4) Indicator box

b. Remove power distribution box cover panel and disconnect both conductors from the voltmeter terminals. Position or isolate disconnected leads so as not to short to each other or to shelter facility ground.

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4.7.1.1 Continuity of conductors.- Continuity resistance which meets the requirements of 3.3.1.1 shall be performed on each power circuit in accordance with the following paragraphs. A calibrated low resistance reading ohmmeter or equivalent shall be used in performing the test.

a. Place the following controls in the ON position.

(1) LIGHT-1 switch

(2) LIGHT-2 switch

(3) Incandescent Cold Start switch

(4) BLOWER-1 switch

(5) BLOWER-2 switch

(6) Main circuit breaker

(7) All branch circuit breakers

b. Place BLACKOUT/BYPASS switch to the BYPASS position.

c. Measure continuity resistance from pin 1 of power connector IN located at the power entrance box to the following: brass (hot) contacts of outside power receptacle J3, Heater-1 power receptacle J4, Convenience power receptacles J5 and J6, Heater-2 power receptacle J7, Blowers -1 and -2 power receptacles J8 and J9, Equipment power receptacles J10 through J14, and Intercom power receptacle J15; black wire termination at each fluorescent lamp DS9 through DS25; black wire termination at each incandescent lamp DS26 through DS33; and pin 1 of power OUT connector J2 at power entrance box.

d. Place all circuit controls listed under a. above in the OFF position except main curcuit breaker.

e. Place blackout/bypass switch to the Blackout position.

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f. Measure continuity resistance from pin 2 of power connector IN to the following: silver (neutral) contacts of outside power receptacle J3, Heater-1 power receptacle J4, Convenience power receptacles J5 and J6, Heater-2 power receptacle J7, Blowers -1 and -2 power receptacles J8 and J9, Equipment power receptacles J10 through J14, and Intercom power receptacle J15; white wire termination at each fluorescent lamp DS9 through DS25; white wire termination at each incandescent lamp DS26 through DS33; and pin 2 of power OUT connector J2.

g. Place main circuit breaker in the OFF position.

h. Measure continuity resistance from pin 4 (ground) of power connector IN to the following: all ground terminals located at the power entrance box, power distribution box, video entrance box, and raceway; green (ground) contacts of outside power receptacle J3, Heater-1 power receptacle J4, Convenience power receptacles J5 and J6, Heater-2 power receptacle J7, Blowers -1 and -2 power receptacles J8 and J9, Equipment power receptacles J10 through J14, and Intercom power receptacle J15; and pin 4 of power OUT connector J2.

i. Resistance measurements made in steps c, f, and h above, shall meet the requirements of 3.3.1.1.

4.7.1.2 Dielectric breakdown voltage.- Dielectric strength tests which meet the requirements of 3.3.1.2 shall be performed on each power circuit in accordance with the following paragraphs. A non-destructive DC dielectric tester shall be used in performing the test.

NOTE: Take preventive measures that will preclude personnel from coming into contact with conductors under test during performance of ALL dielectric and insulation resistance tests. Discharge each conductor to ground following each test, to eliminate danger of shock.

a. Place all circuit controls listed under 4.7.1.1a in the ON position, except LIGHT-1 and LIGHT-2 switches.

b. Place BLACKOUT/BYPASS switch to the bypass position, and lock shelter facility door securely.

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c. Test between pins 1 and 4 of power connector IN. Adjust tester for a value of not less than 500 volts DC output and maintain this potential for 30 seconds. The requirements of 3.3.1.2 shall be met.

d. Place LIGHT-1 and LIGHT-2 switches in the ON position and lock shelter facility door securely.

e. Repeat step c above with tester connected first across pins 1 and 2, and then across pins 2 and 4.

4.7.1.3 Insulation resistance.- Insulation resistance which meets the requirement of 3.3.1.3 shall be made on each power circuit in accordance with the following paragraphs. A megohmmeter or equivalent shall be used in performing the test.

a. Place all circuit controls listed under 4.7.1.1a in the ON position, except LIGHT-1 and LIGHT-2 switches.

b. Place BLACKOUT/BYPASS switch to the BYPASS position, and lock shelter facility door securely.

c. Test between pins 1 and 4 of power connector IN. The output test voltage shall not be less than 100 volts DC nor more than 500 volts DC. Maintain this potential until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. (See Note under 4.7.1.2). The requirements of 3.3.1.3 shall be met.

d. Place LIGHT-1 and LIGHT-2 switches in the ON position and lock shelter facility door securely.

e. Repeat step c above with tester connected first across pins 1 and 2, and then acorss pins 2 and 4.

4.7.2 Power distributional and operational function tests.- Power distributional and operational function tests which meet the requirements of 3.3.2 shall be performed on each power circuit in accordance with the following paragraphs: (Refer to Drawings SC-D-632457 and SC-F-632468).

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4.7.2.1 Test preparation.-

a. Replace or reconnect all electrical loads and line voltmeter removed or disconnected by 4.7.1a and b.

b. Replace power distribution box cover.

c. Place all circuit breakers and switches in the OFF position.

d. Place BLACKOUT/BYPASS switch to the BYPASS position.

e. Place a 115 volt AC test lamp with grounding plug into all convenience, equipment, intercom, and heater receptacles.

f. Open outer exhaust port covers of both blowers.

g. Insert AC line plugs of both blowers into their respective receptacles.

4.7.2.2 Test of line voltmeter and indicator lamp.-

a. Apply 115 volts AC (nominal), 60 cycles, single phase power to shelter facility at power connector IN. Power indicator neon lamp shall light and facility voltmeter at power distribution panel shall indicate power to the shelter.

b. Measure specific line voltage by connecting a calibrated AC voltmeter (0-150 volt scale) between pin 1 and pin 2 of power connector OUT.

c. Read voltage indicated on shelter facility voltmeter. The voltage read shall be within $\pm$ 2% of the voltage measured by step b above.

d. Place main circuit breaker in the ON position.

4.7.2.3 Test of circuit breakers and switches.-

Circuit #1.- (Lights circuit breaker CB1)

a. Place lights circuit breaker CB1 in the ON position. Neon lamp above CB1 shall light.

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b. Place LIGHT-1 switch in the ON position. Half of the roadside and curbside fluorescent lamps (DS14 through DS21) shall light.

c. Place LIGHT-2 switch in the ON position. The remaining roadside and curbside fluorescent lamps (DS9 through DS12 and DS22 through DS25) and front fluorescent lamp(DS13) shall light.

d. Place incandescent cold start switch in the ON position. All incandescent lamps (DS26 through DS33) shall light.

e. Place circuit breaker CB1 in the OFF position. All lamps mentioned in steps a. through d. above shall go out.

f. Replace circuit breaker CB1 in the ON position to restore lights.

g. With shelter door open, place BLACKOUT/BYPASS switch to the BLACKOUT position. All fluorescent and incandescent lamps shall go out.

h. Close shelter door. All fluorescent and incandescent lamps shall light.

i. Place LIGHT-1 switch in the OFF position. Fluorescent lamps mentioned in step b. above shall go out.

j. Place LIGHT-2 switch in the OFF position. Fluorescent lamps mentioned in step c. above shall go out.

k. Place incandescent cold start switch in the OFF position. All incandescent lamps shall go out.

l. Open door and restore lights by placing BLACKOUT/BYPASS switch to BYPASS and LIGHT-1 and LIGHT-2 switches to the ON position.

Circuit #2.- (Blowers circuit breaker CB2)

a. Place blowers circuit breaker CB2 in the ON position. Neon lamp above CB2 shall light.

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b. Place roadside blower switch in the ON position. Roadside blower shall operate and shall exhaust air from the shelter.

c. Repeat step b. above for curbside switch and blower.

d. Place circuit breaker CB2 in the OFF position. Neon lamp above CB2 shall go out and both blowers shall cease to operate.

Circuit #3.- (Heater - 1 circuit breaker CB3)

a. Place Heater-1 circuit breaker CB3 (bottom breaker of tandem) in the ON position. Neon lamp #3 above CB3 and test lamp in Heater-1 receptacle shall light.

b. Place circuit breaker CB3 in the OFF position. All lamps mentioned in step a. above shall go out.

Circuit #4.- (Heater - 2 circuit breaker CB4)

a. Place Heater-2 circuit breaker CB4 (top breaker of tandem) in the ON position. Neon lamp #4 above CB4 shall light and test lamp in heater-2 receptacle shall light.

b. Place circuit breaker CB4 in the OFF position. All lamps mentioned in step a. above shall go out.

Circuit #5.- (Indicator circuit breaker CB5). Refer to Drawing SC-D-632645.

a. Place Indicator circuit breaker CB5 (top breaker of tandem) in the ON position. Neon lamp #5 above CB5 shall light.

b. Depress audio alarm disabled switch of system 1 equipment indicator. The alarm disabled indicator shall show a flashing red indication.

c. Repeat step b. above for systems 2 and 3 equipment indicators.

d. Place circuit breaker CB5 in the OFF position. Neon lamp #5 above CB5 and flashing red indications of systems 1 through 3 equipment indicators shall go out.

e. Place CB5 in the ON position to restore flashing indications.

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f. Depress audio alarm disabled switch of systems 1 thru 3 equipment indicators. The flashing red indications of system 1 thru 3 shall go out.

g. Replace CB5 in the OFF position.

h. Disconnect cable assembly connecting to each indicator box.

i. Connect 24 volts AC across pins E and D of indicator box connector J1 of system 1. Red Traffic Out indication shall light.

j. Repeat step i. for pins C and E. Green Traffic Out indication shall light.

k. Repeat step i. for pins E and G. Red Traffic In indication shall light.

l. Repeat step i. for pins E and H. Green Traffic In indication shall light.

m. Repeat steps i. thru l. for systems 2 and 3 indicator boxes.

n. By use of test jigs, as needed, short across pins K and L of J1 thru J3, systems 1 thru 3. Reconnect cable assembly to system 1 indicator box.

o. Place circuit breaker CB5 in the ON position. Green ANT indications on systems 1 thru 3 shall light.

p. Place CB5 in the OFF position.

q. Repeat steps n. and o. for pins J and L. Red Dummy Load indications shall light.

Circuit #6.- (Convenience circuit breaker CB6)

a. Place convenience circuit breaker CB6 (bottom breaker of tandem) in the ON position. Neon lamp #6 above CB6 and test lamps in outside convenience, inside conveniences (2), and intercom receptacles shall light.

b. Place circuit breaker CB6 in the OFF position. All lamps mentioned in step a. above shall go out.

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Circuit #7.- (Equipment circuit breaker CB7)

a. Place equipment circuit breaker CB7 in the ON position. Neon lamp above CB7 and test lamp in each equipment receptacle shall light.

b. Place circuit breaker CB7 in the OFF position. All lamps mentioned in step a. above shall go out.

Circuit #8.- (Spare circuit breaker CB8)

a. Place spare circuit breaker CB8 in the ON position. Neon lamp above CB8 shall light.

b. Place circuit breaker CB8 in the OFF position. Neon lamp above CB8 shall go out.

Circuit #9.- (Main circuit breaker CB9)

a. Place all circuit breakers CB1 through CB9 and all switches in the ON position. Depress alarm disabled switch of systems 1 through 3 indicators. All neon lamps above circuit breakers shall light. All fluorescent and incandescent lamps shall light. Test lamps in all convenience, outside convenience, intercom, heater-1, heater-2, and equipment receptacles, shall light. Both blowers shall operate. Equipment indicators of systems 1 through 3 shall show a flashing red indication.

b. Place main circuit breaker CB9 in the OFF position. All lamps, blowers and flashing indications mentioned in step a. above shall go out.

c. For remainder of tests, place main circuit breaker and light circuit breaker in the ON position and all other circuit breakers in the OFF position.

4.7.3 Signal (audio) system wiring and components installation tests.- Refer to signal schematic Drawing SC-D-632538. (See 3.3.3)

4.7.3.1 Continuity of conductors.- Continuity resistance which meets the requirements of 3.3.3.1 shall be performed on each audio circuit in accordance with the following paragraphs. A low resistance reading continuity tester or equivalent shall be used in performing the test.

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Test of Equipment Interconnect Harness.- (Refer to Drawing SC-D-632535)

a. Measure continuity resistance between the following pins of cable assemblies marked TRANSMITTER and IND BOX of system 1: a to A, c to B, R to C, P to D, K to E, M to G, and N to H.

b. Measure continuity resistance between cable assemblies marked IND BOX and WG SWITCH of system 1: K to F, J to D, and L to E.

c. Measure continuity resistance between cable assemblies marked TRANSMITTER and WG SWITCH of system 1: W to B, V to A, and U to C.

d. Measure continuity resistance between cable assemblies marked TRANSMITTER and SYSTEM 1 RCVR: H to B and J to A.

e. Measure continuity resistance between the following conductors of assemblies marked TRANSMITTER and CV-425 LINE of system 1: S to black conductor, T to red conductor, and e to shield.

f. Continuity resistance measurements made in steps a through e above shall meet the requirements of 3.3.3.1.

g. Repeat steps a through f above for systems 2 and 3 respectively.

Test of Sighting Cables.- (Refer to Drawing SC-D-632533)

a. Measure continuity resistance between the following pins of SIGHTING -1 connector at antenna entrance box and cable assembly marked RCVR ANT ALIGN-1: A to A, B to B, and C to C.

b. Measure continuity resistance between the following pins of SIGHTING-1 connector and Remote Order Wire binding posts SYS-1 at power entrance box: F to binding post A and G to binding post B.

c. Continuity resistance measurements made in steps a and b above shall meet the requirements of 3.3.3.1.

d. Repeat steps a through c above for systems 2 and 3 respectively.

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Test of Phone/Intercom Lines.-

a. Measure continuity resistance between the phone binding post at power entrance box and the Phone Jack at front wall as follows: binding post A to sleeve of phone jack and binding post B to tip of phone jack. Continuity resistance measurements shall meet the requirements of 3.3.3.1.

b. Repeat step a above for the Intercom binding posts and jack.

Test of Remote Phone Lines.-

a. Measure continuity resistance between the Remote Order Wire binding posts of system 1 and CV-425 LOOP cable assembly of system 1 as follows: binding post A to black conductor and binding post B to red conductor. Continuity resistance measurements shall meet the requirements of 3.3.3.1.

b. Repeat step a above for systems 2 and 3 respectively.

Test of Thru Order Wire Cables.-

a. Using Order Wire Patch cable connect between Order Wire System 1 and Order Wire System 2 of patch panel.

b. Measure continuity resistance between the following pins of cable assemblies marked OW XMTR SYS-1 and OW XMTR SYS-2: A to D, B to E, C to C, D to A, and E to B. Continuity resistance measurements shall meet the requirements of 3.3.3.1.

c. Repeat step b above between systems 1 and 3 with Order Wire Patch cable connected across Order Wire System 1 and Order Wire System 3.

4.7.3.2 Dielectric and insulation resistance tests.- Dielectric and insulation resistance tests which meet the requirements of 3.3.3.2 shall be performed on each audio circuit in accordance with the following paragraphs. A non-destructive dielectric and insulation resistance tester shall be used in performing the test.

Test of Equipment Interconnect Harness.- (Refer to Drawing SC-D-632535)

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a. Connect the following pins of cable assembly marked TRANSMITTER of system 1 to shelter facility ground terminal using appropriate test jigs: a, c, R, P, K, M, N, S, T, W, V, U, H and J.

b. Repeat step a above for pins D, E, and F, of WG SWITCH system 1 cable assembly.

c. Place tester between pin A of cable assembly marked IND BOX of system 1 and shelter facility ground terminal. Remove ground connection from pin a of TRANSMITTER system 1 cable assembly.

d. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. The requirements of 3.3.3.2 shall be met. Re-establish ground connection to the conductor under test.

e. Repeat steps c and d above for the following pins of IND BOX system 1 cable assembly: B, C, D, E, G, H, L, K, J, and V; for the following pins of WG SWITCH system 1 cable assembly: A, B, and C; and for SYSTEM 1 RCVR cable assembly: A and B; each time ungrounding the pins correlating with the circuit being tested and reconnecting to shelter facility ground terminal following completion of test.

f. Repeat steps a through e above for systems 2 and 3.

Test of Sighting Cables.-

a. Place tester between pin A of SIGHTING-1 connector at antenna entrance box and shelter facility ground terminal.

b. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. The requirements of 3.3.3.2 shall be met.

c. Repeat steps a and b above for pins B, C, F, and G of SIGHTING-1 connector.

d. Repeat steps a through c above for SIGHTING-2 and SIGHTING-3 connectors.

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Test of Remote Order Wire, Intercom, and Phone lines.-

a. Place tester between the Remote Order Wire SYS-1 binding post A and all other binding posts connected to shelter facility ground.

b. Adjust tester to 500 volts DC and maintain this potential for a continuous period of not less than 5 seconds. The requirements of 3.3.3.2 shall be met.

c. Repeat steps a and b above for each remaining Remote Order Wire, Intercom and Phone binding post.

Test of Thru Order Wire Cables.-

a. Using Order Wire Patch cable connect between Order Wire System 1 and Order Wire System 2 of patch panel.

b. Connect the following pins of cable assembly marked OW XMTR SYS-1 to Shelter Facility ground terminal: A, B, C, D, and E.

c. Place tester between pin A of cable assembly marked OW XMTR SYS-2 and shelter facility ground terminal. Remove ground connection from pin D of OW XMTR SYS-1 cable assembly.

d. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. The requirements of 3.3.3.2 shall be met. Re-establish ground connection to the conductor under test.

e. Repeat steps c and d above for the following pins of cable assembly OW XMTR SYS-2: B, C, D, and E; each time ungrounding the pin of cable assembly OW XMTR SYS-1 correlating with the circuit being tested as follows: E, C, A, and B respectively. After each test reconnect the pin being tested to shelter facility ground terminal.

f. Repeat steps b through e above substituting system 3 for system 2 and connecting Order Wire Patch cable across Order Wire System 1 and Order Wire System 3 of patch panel.

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4.7.4 Video and radio frequency (RF) system wiring and components installation.- (See 3.3.4)

4.7.4.1 Continuity of conductors.- Continuity resistance which meets the requirements of 3.3.4.1 shall be performed on each video and RF circuit in accordance with the following paragraphs. A calibrated low resistance reading ohmmeter or equivalent shall be used in performing the test.

a. At CABLE/RADIO TERMINAL patch panel make the following cable connections using video patch cords:

VIDEO CABLE	TO	VIDEO EQUIPMENT
IN SYS-1		IN SYS-1
OUT SYS-1		OUT SYS-1
IN SYS-2		IN SYS-2
OUT SYS-2		OUT SYS-2
IN SYS-3		IN SYS-3
OUT SYS-3		OUT SYS-3

b. At Radio Repeater patch panel make the following cable connections using video patch cords: IN SYS-1 to OUT SYS-1; IN SYS-2 to OUT SYS-2; and IN SYS-3 to OUT SYS-3.

c. Circuit continuity resistance which meets the requirements of 3.3.4.1 shall exist when tested in accordance with Table V. The method of testing shall be inner connector to inner conductor and outer conductor (shield) to outer conductor (shield).

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TABLE V
Continuity Measurements

FROM	TO
FROM CABLE SYS-1	VID ENT BOX SYS-1 (Male)
TO CABLE SYS-1	VID ENT BOX SYS-1 (Female)
FROM CABLE SYS-2	VID ENT BOX SYS-2 (Male)
TO CABLE SYS-2	VID ENT BOX SYS-2 (Female)
FROM CABLE SYS-3	VID ENT BOX SYS-3 (Male)
TO CABLE SYS-3	VID ENT BOX SYS-3 (Female)
REPEATER IN SYS-1	REPEATER OUT SYS-1
REPEATER IN SYS-2	REPEATER OUT SYS-2
REPEATER IN SYS-3	REPEATER OUT SYS-3
TEST OUT SYS-1	DIR COUPLER SYS-1
TEST OUT SYS-2	DIR COUPLER SYS-2
TEST OUT SYS-3	DIR COUPLER SYS-3
TRAFFIC SYS-1 (XMTR)	TRAFFIC OUT SYS-1 (RCVR)
TRAFFIC SYS-2 (XMTR)	TRAFFIC OUT SYS-2 (RCVR)
TRAFFIC SYS-3 (XMTR)	TRAFFIC OUT SYS-3 (RCVR)

4.7.4.2 Dielectric and insulation resistance tests.- Dielectric and insulation resistance tests which meet the requirements of 3.3.4.2 shall be performed on each video and RF circuit in accordance with the following paragraphs. A non-destructive dielectric and insulation resistance tester shall be used in performing the test.

a. Repeat cable connections under 4.7.4.1a.

b. Remove all lightning protector elements at video entrance box.

c. Apply test potential of 1500 volts DC for not less than 30 seconds between the following pin connections and terminal ground located at video entrance box. The requirements of 3.3.4.2 shall be met.

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(1) System-1 male

(2) System-1 Female

(3) System-2 male

(4) System-2 female

(5) System-3 male

(6) System-3 female

d. Replace lightning protector elements removed by step b above.

e. Repeat step c above applying test potential of 1100 volts DC.

f. Apply test potential of 1500 volts DC for not less than 5 seconds between inner conductor and outer conductor (shield) of the following cable assemblies: REPEATER IN SYSTEM-1, REPEATER IN SYS-2, REPEATER IN SYS-3, TEST OUT SYS-1, TEST OUT SYS-2, TEST OUT SYS-3, TRAFFIC SYS-1(XMTR), TRAFFIC SYS-2(XMTR), and TRAFFIC SYS-3(XMTR). The requirements of 3.3.4.2 shall be met.

4.7.4.3 Insertion loss.- Insertion loss which meets the requirements of 3.3.4.3 shall be performed on each waveguide transmission line in accordance with the following paragraphs. A system such as shown in figure 1 for a frequency range between 4.4 to 5.0 GHz shall be used. The following calibrated test equipment, or equal, shall be provided by the contractor in the quantities necessary to accomplish the testing specified herein:

Oscillator, Sweep, Mainframe HP 8690B
Module, RF HP 86341B
Drawer, RF HP 8700A
Adapter, Coax/Waveguide HP G281A
Coupler, Directional, 10 DB HP G752C
Coupler, Directional, 3 DB HP G752A
Termination, 50 ohm HP G910A
Mount, Thermistor HP G486
Meter, Power HP 432A
Amplifier, Leveler HP 8404A
Calibrator, Power Meter HP 8477A
Voltmeter, Digital HP 3440A
Recorder, X-Y HP 7035B
Ancillary components

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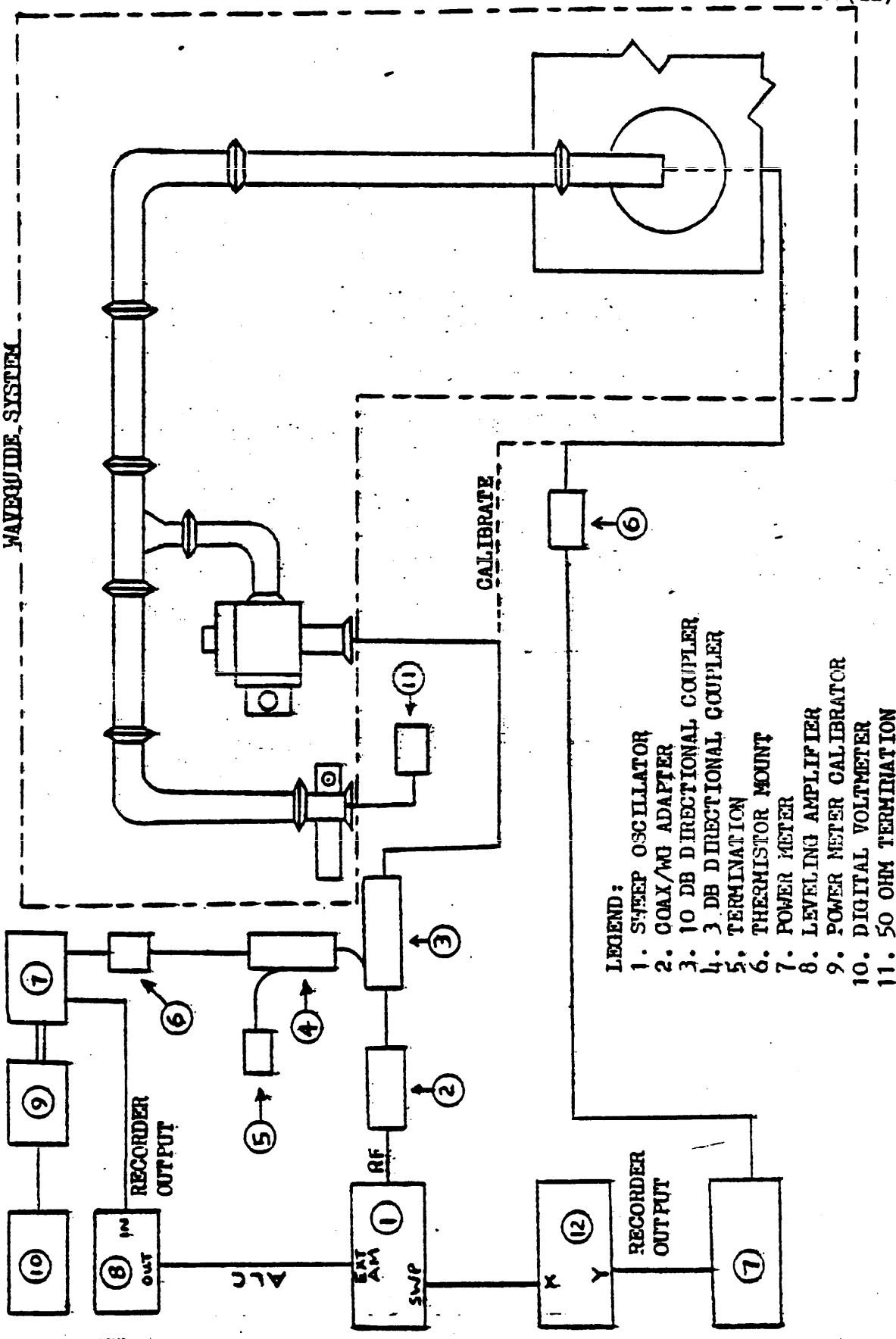


FIG. 1 TYPICAL INSERTION LOSS SETUP

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a. Set sweep oscillator for automatic sweep between 4.4 and 5.0 GHz.

b. Connect equipment for calibration purposes as shown in figure 1.

c. Plot a zero db reference line on the X-Y recorder with power meter #1 leveled at 0.5 mw RF power and calibrator off, by following manufacturer's suggested procedures.

d. Accurately calibrate the X-Y recorder attenuation span by use of the digital voltmeter and plot additional reference lines at 0.25 db, 0.5 db, 1.0 db, and 1.25 db by applying accurately known values of dc power to power meter #1 with the calibrator.

e. Turn off calibrator and insert the test item as shown in figure 1.

f. Sweep the required frequency range and obtain an insertion loss (db) versus frequency (GHz) plot . The insertion loss as obtained on the X-Y recorder graph shall not exceed 1.25 db throughout the required frequency range.

g. Repeat steps a thru f above placing source end at system-1 antenna end and load end at system-1 receiver end, terminating system-1 transmitter end with a 50 ohm load.

h. Repeat steps a thru g for systems 2 and 3 respectively.

4.7.4.4 Voltage standing wave ratio (VSWR).- VSWR which meets the requirements of 3.3.4.4 shall be performed on each waveguide transmission line in accordance with the following paragraphs. A system such as shown in figure 2 for a frequency range between 4.4 to 5.0 gigahertz shall be used. The following calibrated test equipment, or equal, shall be provided by the contractor in the quantities necessary to accomplish the testing specified herein:

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Oscillator, Sweep, Mainframe HP 8690B
Module, RF HP 86341B
Drawer, RF HP 8700A
Adapter, Coax/Waveguide HP G281A
Coupler, Directional, 10 DB HP G752C
Detector, Crystal HP G424A
Detector, Crystal w/sq law load HP G424 w/opt 02
Termination, 50 ohm HP G910A
Short, HP G920A
Oscilloscope, Mainframe HP 140A
Plug-in unit HP 1416A
Recorder, X-Y HP 7035B
Ancillary components

a. Make necessary focus, intensity and dc balance adjustements to the oscilloscope in accordance with the manufacturer's guidelines. Set oscilloscope to accept dc inputs. Set Mode to LOG, Sensitivity to 5 db/cm, and position vertically the trace at the bottom graticule.

b. Set sweep oscillator for automatic sweep between 4.4 and 5.0 gigahertz.

c. Connect equipment for calibration purposes as shown in figure 2.

d. Calibrate oscilloscope for direct return loss (db) readout in accordance with the manufacturer's procedures.

e. With recorder turned OFF, connect the X and Y inputs to the oscilloscope's HORIZ and VERTZ outputs. Set sweep oscillator to manual sweep and manual sweep control fully counterclockwise.

f. Set oscilloscope mode to CAL(db)-0 and sensitivity to 10 db/cm.

g. Set X-Y recorder vertical range to 100 mv/inch; and variable horizontal range to obtain full scale deflection.

h. Turn on the X-Y recorder and position the pen over the desired 0db (vertical) and start frequency (4.4 GHz) reference point on the graph paper using the recorder's zero controls.

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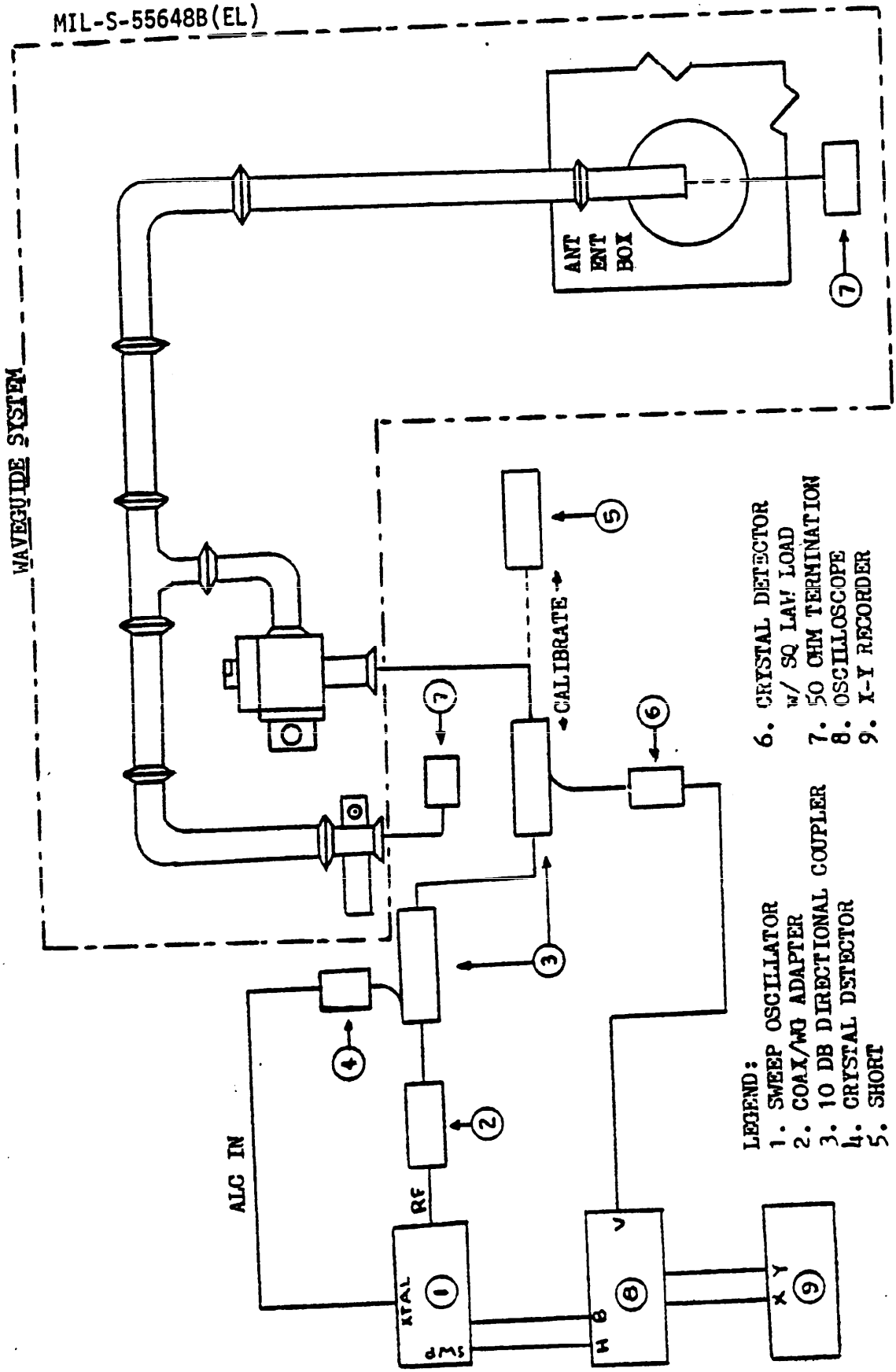


FIG. 2 TYPICAL VSWR TEST SETUP

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i. Set manual sweep control of sweep oscillator to fully clockwise and adjust oscilloscope horizontal amplitude to position pen to desired stop-frequency (5.0 reference). Check and readjust zero at start frequency reference if necessary. The recorder horizontal span is now calibrated.

j. Set oscilloscope mode to -10, -20, and -30 db successively and adjust oscilloscope recorder output vertical amplitude so pen moves desired number of vertical divisions on the graph paper. Check and readjust vertical zero if necessary.

k. Set oscilloscope mode to LOG. The recorder vertical span is now calibrated in db/division corresponding to the oscilloscope sensitivity.

l. Insert system 1 waveguide system as shown in figure 2. Place waveguide switch to "ANT" position.

m. Read return loss (db) directly from the calibrated display.

n. Set the trigger control of the sweep oscillator to AUTO. Set sweep time for about 20-30 seconds.

o. Obtain an X-Y plot of the return loss by pushing the time button on the sweep oscillator.

p. Calculate VSWR on the plot at the worst condition(s) by the use of the following formula. The requirements of 3.3.4.4 shall be met.

$$\log_{10} \rho = \frac{-\text{Return loss (db)}}{20}$$

$$\text{VSWR} = \frac{(1+\rho)}{(1-\rho)}$$

where ρ is the magnitude of reflection coefficient.

q. Repeat VSWR test of system 1 waveguide with source connected first at the RCVR end and then at the ANT end.

r. Repeat VSWR tests of steps a thru q above for systems 2 and 3.

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4.7.4.5 Isolation.- When tested as specified in the following paragraphs the isolation of each completed waveguide transmission line shall meet the requirements of 3.3.4.5. A system such as shown in figure 3 for a frequency range between 4.4 to 5.0 gigahertz shall be used. The following calibrated test equipment, or equal, shall be provided by the contractor in the quantities necessary to accomplish the testing specified herein:

Oscillator, Sweep, Mainframe HP 8690B
Module, RF HP 86341B
Drawer, RF HP 8700A
Adapter, Coax/Waveguide HP G281A
Coupler, Directional, 10 DB HP G752C
Attenuator, Variable, Precision HP G302A
Detector, Crystal HP G424A
Indicator, Standing Wave HP 415E
Short, Sliding HP G923A
Load, Sliding HP G914A
X-Y Recorder HP 7035B
Ancillary Components

a. Set sweep oscillator for automatic sweep between 4.4 and 5.0 GHz, with the sweeper output amplitude modulated at 1 KHz.

b. Connect system 1 waveguide system as shown in figure 3 for calibration purposes.

c. Plot calibrated grid lines on the recorder at variable attenuator settings of 20 DB, 25 DB, 30 DB and 40 DB, respectively. Note: Rapidly phase the short during each calibration sweep.

d. Return variable attenuator to zero and replace the short with a degraded sliding load as shown in the figure. (Follow the manufacturer's instruction for degrading the load).

e. Set sweep time for about 40 seconds.

f. Trigger the sweep oscillator to obtain a plot of isolation (db) versus frequency (GHz), constantly phasing the load back and forth during the entire sweep.

g. The isolation obtained by the plot shall meet the requirements of 3.3.4.5.

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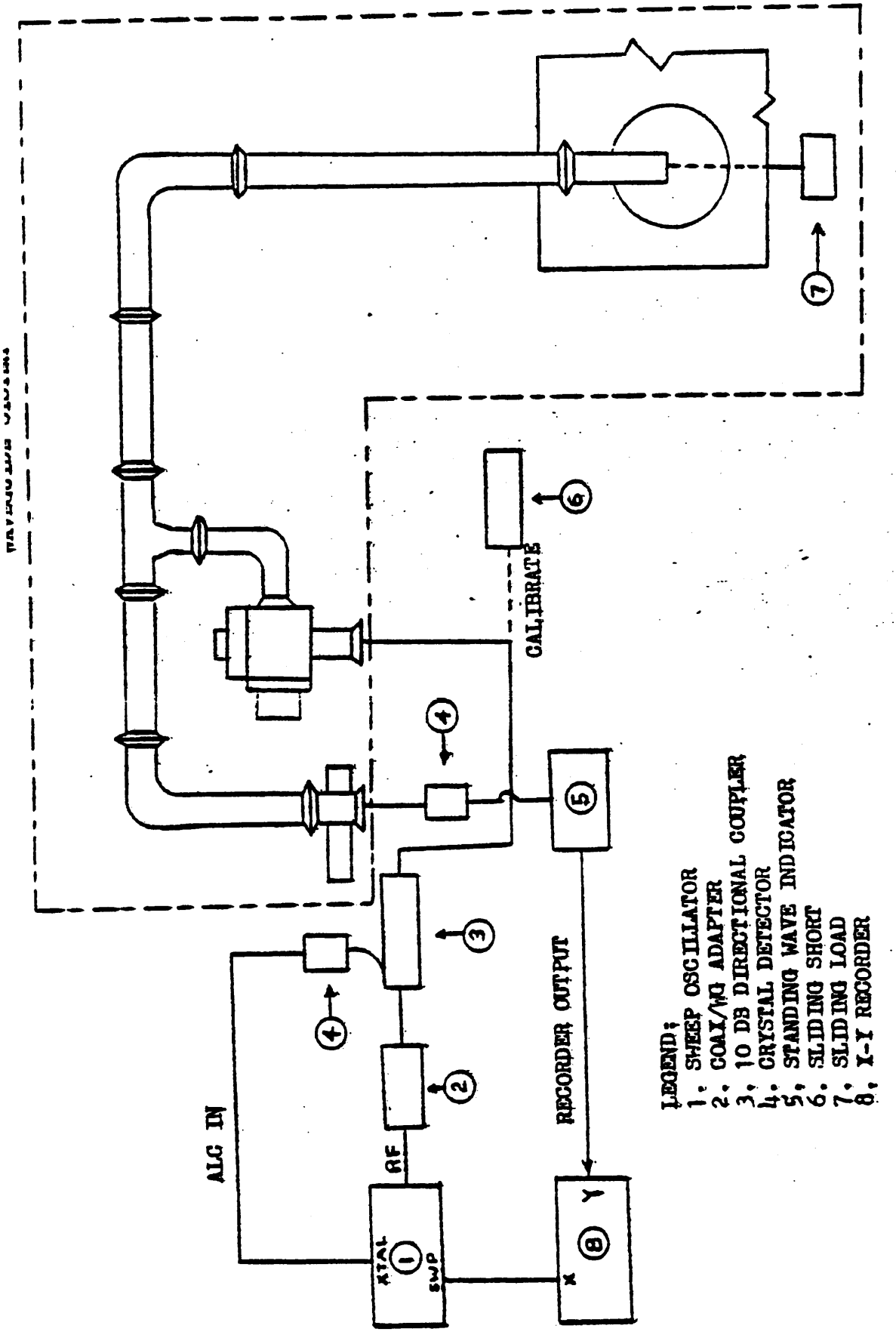


FIG. 3 TYPICAL ISOLATION TEST SETUP

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h. Repeat system 1 waveguide isolation tests, steps a thru g, with source connected at antenna end, sliding load at receiver end, and test measurements at transmitter end.

i. Repeat isolation tests of steps a thru h above for systems 2 and 3.

4.8 Systems operation tests.- Install all operational equipments in their respective locations and connect power, signal, and RF cable assemblies to their designated receptacles and connectors. (See 3.4) (See figure 4 for block diagram).

a. Preliminary setup.-

(1) At CABLE/RADIO TERMINAL patch panel make the following cable connections using video patch cords:

VIDEO CABLE	TO	VIDEO EQUIPMENT
IN SYS-1		IN SYS-1
OUT SYS-1		OUT SYS-1
IN SYS-2		IN SYS-2
OUT SYS-2		OUT SYS-2

(2) Position multiplexers TD-353/U and TD-204/U to facilitate connections to the video entrance box connectors of systems 1 and 2.

(3) Make the following cable connections between the multiplexers of system 1 using Cable Assemblies CG-1040B/U (5 ft).

TD-353/U	TO	TD-204/U
PCM IN		PCM OUT
PCM OUT		PCM IN
TIM IN		TIM OUT
TIM OUT		TIM IN

(4) Connect one end of cable assemblies CG-2437/TCC (15 ft) and CG-2438/TCC (15 ft) to the FROM CABLE and TO CABLE connectors of the TD-204/U system 1 respectively.

(5) Connect other end of cable assemblies CG-2437/TCC and CG-2438/TCC to cable assembly CX-10734( )/G.

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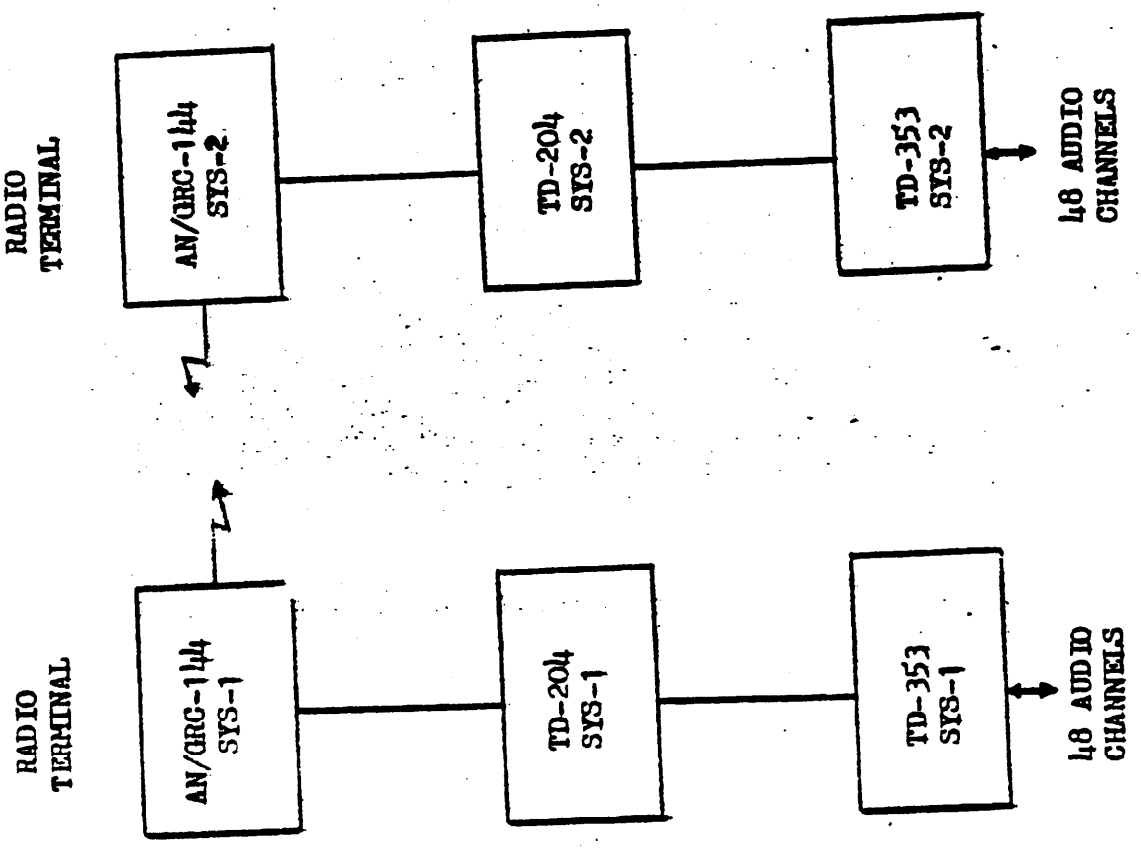


FIG. 4 SYSTEM TEST SETUP

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(6) Connect the remaining end of cable assembly CX-10734( )/G to SYS-1 connector at video entrance box.

(7) Connect Headset H-91A/U to system 1 TD-204.

(8) Repeat steps a(1) thru a(7) above for system 2 multiplexers and system 2 connector at video entrance box.

(9) Connect waveguides CG-3307/GRC (15 ft) to system 1 and system 2 antenna connectors at antenna entrance box. Separate the open ends to a distance of about 5 feet.

(10) Set 2 wire/4 wire switch of each TD-353/U to 2 wire; and MASTER/SLAVE switch to MASTER.

(11) Set TRAFFIC SEL switch of each TD-204/U to 48 AR: READ/ZERO SET/NORM OPR switch to NORM OPR; and the MILE switches on panels 6A4 and 6A5 to 1/2.

(12) Set multiplexers for proper operation in accordance with the instruction card on the equipment.

b. Preliminary starting procedures.- The following procedures apply to the systems under test. Use 4.6 GHz as SYS-1 receive and SYS-2 transmit frequencies; and 4.8 as SYS-1 transmitter and SYS-2 receiver frequencies.

(1) At radio patch panel of transmitter, insert plug into jack J13 for cable terminal operation.

(2) Place waveguide switch SA-1679/GRC in the TO LOAD (TEST) position.

(3) Preset the Receiver R-1467/GRC-144 controls as follows:

(a) Meter selector switch: OFF.

(b) Preselector bandpass filter: Required receiver frequency.

(c) Post selector bandpass filter: Required receiver frequency.

(d) Local oscillator bandpass filter: Required receiver frequency.

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(e) 70 MHz IF amplifier: AGC SEL to DIV OFF.

(f) Electrical frequency synthesizer: Required receiver frequency.

(4) Preset the Transmitter T-1054/GRC-144 controls as follows:

(a) Meter selector switch: OFF (TRANSMIT).

(b) Radio test set: OUTPUT LEVEL TO LOW.

(c) 100 MHz radio freq osc: ON/OFF switch to OFF.

(d) Attenuator assembly: 48 CH/96 CH to 48 CH.

(e) Radio freq bandpass filter: Required transmitter frequency.

(f) Digital data modem: ALARM TRAFFIC SELECT 48 CHAN/96 CHAN switch to 48 channel position and CABLE LOOP TEST NORMAL/TEST switch to NORMAL.

(g) Orderwire assembly: Module Test selector switch to position 1.

(h) Electrical freq synthesizer: Required transmitter rrequency.

c. Starting procedures.—

(1) Place POWER ON-OFF switch on meter panel of transmitter to ON, and check that the following conditions exist:

(a) 5 AMP SLO-BLOW neon indicator on meter panel shall not light.

(b) All power supply indicator lamps shall light.

(c) The 5V and 28V indicator lamps on electrical frequency synthesizer shall light.

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(d) All fault indicator lamps on electrical frequency synthesizer shall not light. Press the TEST pushbutton switch to check that all fault indicator lamps light.

(e) Correct indication mode indicator (Cable terminal operation or Radio Repeater operation) lamp on radio patch panel shall light.

(2) Place POWER ON-OFF switch on meter panel of receiver to ON, and check that the following conditions exist:

(a) The SLO-BLOW 5 AMP and TDA 1 AMP indicators on meter panel shall not light.

(b) All power supply indicator lamps shall light.

(c) The 5V and 28V indicator lamps on electrical frequency synthesizer shall light.

(d) All fault indicator lamps on electrical frequency synthesizer shall not light. Press the TEST pushbutton switch to check that all fault indicator lamps light.

d. Establishing initial radio communications.-

(1) Set Waveguide Switch SA-1679/GRC to the TO ANT. (NORMAL) position of systems 1 and 2.

(2) Assure that the multiplexers and radios are functioning properly. NOTE: The unterminated waveguides may be moved accordingly to receive a good signal strength.

(3) Check that the following conditions exist on systems 1 and 2 transmitter.

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Indicator	Color Lighted
RADIO TEST SET	Green
WAVEGUIDE SW	Green
RING	Green or amber
CNTRL ALARM	Green
TRAF	Green
RF POWER	Green
SYNTH LOCK	Green
REFL RF POWER	Green
CABLE TO RADIO MODEM	Green
RADIO TO CABLE MODEM	Green

(4) Check that the following conditions exist on systems 1 and 2 receivers.

Indicator	Color Lighted
CARR (IF)	Green
SPARE	Green
SYNTH LOCK	Green
TRAF	Green

(5) Check that the following conditions exist on systems 1 and 2 indicator boxes:

Indicator	Color Lighted
TRAF IN	Green
TRAF OUT	Green
ANT	Green

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(6) Perform the following transmitter (systems 1 & 2) metering checks using the meter and meter selector switch on meter panel. The chart below lists the metering positions to be checked and the required meter indications.

MODULE TEST Selector switch position	MODULE TEST Indicator lamp	CNTRL ALARM Indicator lamp	Loudspeaker tone output
1	OFF	Green	None
1 (press the MODULE TEST indicator lamp) pushbutton switch	ON	Green	None
2	OFF	Green	4KHz
3	OFF	Green	4KHz
4*	OFF	Green	4KHz
5	OFF	Green	4KHz
6	OFF	Green	4KHz
7	OFF	Green	4KHz
8	OFF	Green	4KHz
9	OFF	Green	1.6KHz
10	OFF	Green	1.1KHz
11	OFF	Green	Alternating 1.1 and 1.6KHz
12	ON	RED	None
12 (momentarily press RESET pushbutton switch. Return to 1	OFF	Green	None

*MODULE TEST and CNTRL ALARM indicator lamps will be lighted red if pcm traffic is not present.

e. Operational tests.- Prior to checking the following procedures, check to see that all equipments are operating properly.

(1) Press the RING pushbutton on transmitter, system 1, meter panel. The RING indicator directly above the pushbutton will change from green to amber each time the RING pushbutton is depressed and a 1.6 KHz tone signal will be heard over the loudspeaker mounted on the meter panel of system 1 transmitter.

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(2) At system 2 transmitter, the RING indicator on meter panel will change from green to amber, and a 1.6 KHz tone will be heard over the loudspeaker. The ring signal will be heard at the two external multiplexers TD-204/U.

(3) Communicate over all phones connected to the radios and multiplexers TD-204/U. Orderwire communications will also be heard over the loudspeakers.

(4) Repeat above procedures ringing from system 2 transmitter.

(5) Ring TD-204/U system 1 from TD-204/U system 2. Ringing signals should also be received at both radio sets.

(6) Repeat step (5) ringing TD-204/U system 2 from TD-204/U system 1.

(7) Connect Telephone Sets TA-312/PT to the remote orderwire of systems 1 and 2. Set CV-425's of systems 1 and 2 for proper operation.

(8) Ring TA-312/PT of system 1 from TA-312/PT of system 2; ring TA-312/PT of system 2 from TA-312/PT of system 1; establish communications between the two phones.

(9) Connect Headsets H-91A/U to each TD-353/U. Establish communications between the two headsets using 12 random channels between 1-48.

(10) Repeat all systems operation tests, steps a thru e, substituting system 3 for system 1.

4.9 Vehicular transportation test.- The facility shall be secured (and blocked to prevent shifting) to a government furnished 2-1/2 ton, M-35 truck and transported over the specially prepared course described in 4.9.1 for 10 laps at a speed of 5 mph, 10 laps at a speed of 10 mph, 10 laps at a speed of 15 mph, and 10 laps at a speed of 18-20 mph. A lap is defined as traversing the course in one direction. The facility shall meet the requirements of paragraph 3.5 after completion of the test.

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NOTE: The facility shall be loaded to simulate the AN/TRC-138( ) using live and dummy loads. The exhaust blowers and housing shall be mounted outside the shelter facility at its normally operating position. (See 6.2b for this test)

4.9.1 Vehicular transportation course.- The course for the military truck transportation shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have a 1 x 1-inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be anchored securely by any convenient means. The ground on which the timbers are placed shall be hard and level.

4.10 Rail transport, impact test.- The shelter facility loaded as specified in paragraph 4.9, secured to a flatcar (the test flatcar) and coupled to two other flatcars shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter facility tiedown slings, shall be as specified in figure 5 and shall meet the provisions of the test as specified in paragraph 4.10.1. The test flatcar (less the shelter facility) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in paragraph 4.10.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the government technical representative) shall be impacted against the stationary loaded test car coupled to the two empty flatcars all with brakes off. The test shall consist of four impacts. Two impacts with the shelter facility positioned longitudinally on the flatcar, one from each end of the shelter facility and two impacts with the shelter facility positioned transversely on the flatcar, one from each side of the shelter facility. The lashing, blocking and wedges holding the shelter facility in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter facility shall meet the requirements of paragraph 3.6.

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4.10.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles, this shall be accomplished by means of power sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings, along the full length of the castings. These four blocking members shall not be notched or shimmed to contact the trim along the sides of the shelter. The 4 x 4 and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in the figure referred above. All other blocking shall be firmly secured as shown in the figure. Lashing shall consist of four wire rope slings, each including 3/8 inch diameter 6 x 19, flexible steel, preformed wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckle, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9000 pounds minimum. Before each impact, each sling shall be taut (600 - 800 pounds tension).

4.10.2 Loaded gondola car.- The load in the loaded gondola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, crushed concrete or metal scrap (other than sheet metal, metal shavings or other springly forms of metal).

b. Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

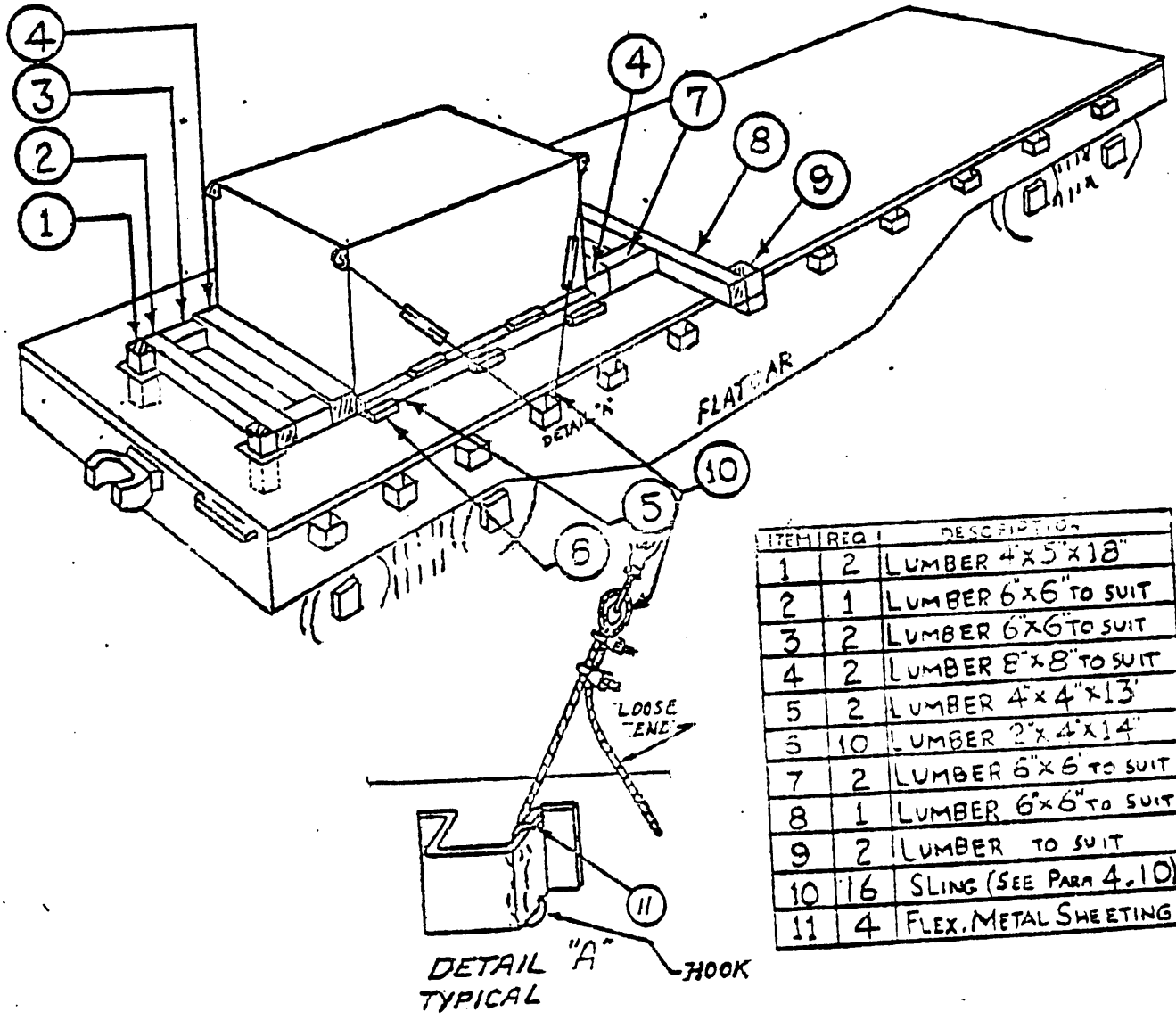
4.11 Drop tests.-

4.11.1 Flat drop.- The shelter facility loaded as in 4.9, shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of 3.7.

4.11.2 Rotational drop.- The shelter facility loaded as in 4.9 shall have one skid resting on a nominal 4 inch high timber. The outer edge of the opposite skid shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of 3.7.

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FIGURE 5



ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"x5"x18"
2	1	LUMBER 6"x6" TO SUIT
3	2	LUMBER 6"x6" TO SUIT
4	2	LUMBER 8"x8" TO SUIT
5	2	LUMBER 4"x4"x13'
6	10	LUMBER 2"x4"x14"
7	2	LUMBER 6"x6" TO SUIT
8	1	LUMBER 6"x6" TO SUIT
9	2	LUMBER TO SUIT
10	16	SLING (SEE PARA 4.10)
11	4	FLEX. METAL SHEETING

NOTES:
a. The blocking lumber shall be notched to accomodate towing eyes, bolts and rivet heads.
b. The blocking lumber shall be rigidly fastened to the floor of the flatcar.

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4.12 Watertightness test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc., shall be subjected to the following two part test: (NOTE: Perform tests with inside cover of entrance box removed).

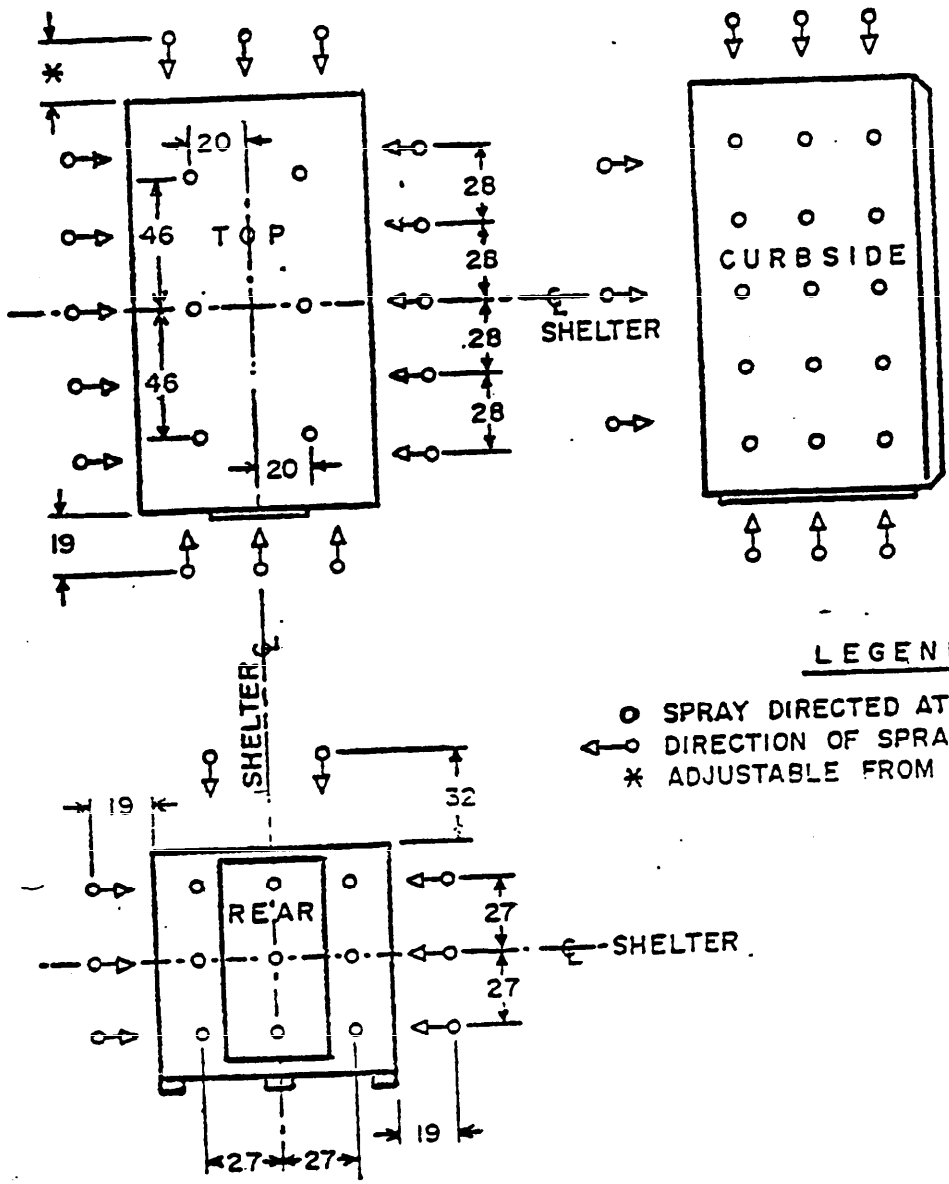
a. Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Company, Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches.

All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. (See figure 6 for nozzle location). Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of 4.13, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door-end panel). Upon completion of the above test, the shelter facility shall meet the requirements of 3.8. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination the holes shall be sealed with rivets and sealer.

b. Remove exhaust blowers and housing from its operating position to its transit position inside the shelter facility and secure openings with the exhaust blower covers. Open all other covers to operating position (with exception of the door vent panel at the rear of the shelter) and repeat a above.

4.13 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 21 inches (measured from the bottom of the skids) and meet the requirements of 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tiedowns. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

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NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 6

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4.14 Mounting insert tests.-

4.14.1 First article and group C insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table VI. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

4.14.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Each selected insert shall be tested in accordance with Table VI. After test, the requirements of 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

TABLE VI - Mounting Inserts Loads

Insert Size	Preproduction Test		Production Test	
	Axial(lbs)	Torque(in lbs)	Axial(lbs)	Torque(in lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

NOTE: Torque loads applied through bolt whose head holds a stationary plate against head of insert.

4.15 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.20.4.

4.16 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

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a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks, patch panels, and cabinets.

4.17 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.22 and Table VII.

TABLE VII - Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated.
	2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible.
	3. Welds - cracked or porous.
	4. Sealer improperly applied (holes, separations, or lack of adhesion).
	5. Mounting inserts missing or mislocated.
	6. Burrs or sharp edges in wire duct not removed.
	7. Equipments or bracketry loose or missing.
	8. Doors, etc., inoperative.
	9. Electrical receptacles, switches, jacks, connectors, wire ducts or other electrical components loose, improperly located or inoperative.
	10. Locking or holding devices missing or inoperative.
	11. Gaps between sections of wire duct exceed 1/16".
Minor	1. Doors not easily operable.
	2. Fastening devices difficult to operate.
	3. Finish - abrasions or scratches.
	4. Burrs and sharp edges on bracket not removed.
	5. Screws loose.
	6. Gaskets, not completely and securely adhered.

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4.18 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.19 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.7 Power, Signal, and Video Wiring

Paragraph 4.17 Visual and Mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the shelter facility shall be in accordance with the applicable requirements of Specification MIL-S-55505.

6. NOTES

6.1 Intended use.- The Shelter, Electrical Equipment S-393( )/TRC-138 is the shelter facility portion of the AN/TRC-138( ) and it is used to house the operational equipments including:

a. 3 ea AN/GRC-144 Radio Set

b. 3 ea CV-425( )/U Converter, Telephone Signal

c. 3 ea H-156( )/U Handset

d. 3 ea TA-312( )/PT Telephone

e. 1 ea LS-147( )/FI Intercom

forming the Radio Repeater Set.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

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b. First Article and Group C inspection of the shelter facility requires use of contractor fabricated dummy loads for the government furnished live loads (radio components) unless otherwise specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-393( )/TRC-138.

d. A list of equipments to be government loaned for testing purposes.

e. Submission of the statement referenced in 3.18.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally non-destructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.5)

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-393B/TRC-138. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited to severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army - EL

Preparing Activity:
Army - EL

Project No. 5820-A563

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-55602B(EL) SHELTER, ELECTRICAL EQUIPMENT S-393( )/TRC-139

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

___ YES ___ NO *(If "yes"; in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

ESC-FM 1068-68

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Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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